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Jeff Shen

✉: jshenschool@gmail.com
☎: +1 437-345-1552

📄: github.com/al-jshen
in: [linkedin.com/in/jeff-shen](https://www.linkedin.com/in/jeff-shen)

Education

H.B.Sc., Mathematics and Statistics
University of Toronto, Toronto, ON

2018-2022

Technical Skills

Data: Python, NumPy/SciPy, Pandas, PyTorch, Scikit-Learn, Matplotlib, Seaborn, SQL, NoSQL, R, Excel
Other: shell scripting, vim, version control (git), cloud computing (AWS, Azure, Google Cloud Platform), Unix/Linux, SymPy, Astropy, Selenium, NLTK, $\text{\LaTeX}$, regular expressions

Relevant Coursework

2019 Fall	Stars and Planets
2019 Fall	Probability and Statistics I
2019 Fall	Ordinary Differential Equations
2019 Summer	Multivariable Calculus
2019 Winter	Introduction to Computer Science
2018 - 2019	Calculus
2018 - 2019	Linear Algebra I, II
2018 Fall	An Introduction to Statistical Reasoning and Data Science

Independent Coursework

2019	Data-Driven Astronomy , The University of Sydney (Coursera) Pulsars, black holes, exoplanets, stellar evolution, distances, redshift, galaxy classification. Image stacking, algorithm analysis, crossmatching, databases, regression, classification, machine learning. Implementation in Python with Astropy; SQL.
2019	Intro to Deep Learning with PyTorch , Udacity Mathematics of neural networks. Data preparation, regularization techniques. Image classification with CNNs and OpenCV. Time series with RNNs. Implementation in Python with PyTorch.
2018	Intro to Machine Learning , Udacity Classifiers (naïve Bayes, SVMs, ensemble methods), data processing, feature selection, regression, clustering, outlier detection, PCA, evaluation metrics. Implementation in Python with scikit-learn.

Projects

2019	Image classification with CNNs Cleaned a dataset of several thousands of images of flowers. Optimized a deep learning model in Python and applied cutting-edge techniques to achieve 99% accuracy in the classification of species of flowers. Could potentially be adapted for classification of astronomical objects.
2018	Analysis of global Internet usage statistics Spearheaded the programming aspect of a collaborative statistical analysis project. Proposed research questions and used R to develop data-backed answers. Gave a presentation of results.

Volunteering

2015 - 2018	Python Workshop Lead , Sir Winston Churchill Secondary, Vancouver, B.C. Coordinated and delivered workshops to prepare students for national programming contest.
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Outreach

2019 - Present	Space Science Journalist (Current in Space segment) , <i>The Star Spot</i> podcast Write and record research-focused stories about the latest news in astronomy.
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Interests

Languages: English (native), French, Chinese, Norwegian (basic)
Other: *nix customization, urban beekeeping, baking, reading, computer programming